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Estimation of Mass and Center of Gravity of Graspless Unknown Object Using Gravity Equi-Effect Planes

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Abstract

In many cases of manipulating an object stably and accurately by robot, it is required to know the center of gravity and mass of the object. For the case when the weight or shape of an object is over the grasp capacity of a robot hand, this paper proposes a technique that can estimate the center of gravity and mass of the graspless and shape-unknown object. A plane called Gravity Equi-Effect Plane is first defined, which contains the center of gravity and a contact line where the object is in line-contact with an environment. If three or over three orientation-different gravity equi-effect planes of an object are obtained, the center of gravity of the object can be estimated by the intersect point of the planes. In order to estimate the gravity equi-effect plane, Tip Operation by robot finger, which tips the object repeatedly, is proposed. Then an algorithm to estimate the gravity equi-effect plane and an algorithm to estimate the mass and center of gravity of the object are addressed by using the fingertip position and force information measured from tip operations.

1 Introduction

In order to move or operate a heavy and/or big object well, people usually take advantage of the contact between the object and its environment. In the same way, this operation can be applied to robot manipulation, where the operation is called as *Graspless Manipulation* since the object is hard to be firmly grasped and wholly held up. The robot pushing operation [5] [6] [7], robot tumbling operation [12], robot pivoting operation [8], and some assembly operations [9], for example, are the graspless manipulations.

For manipulating an object stably and accurately by robot, it is required to know the mass and center of gravity of the object. To the contrary, it will be difficulty to make an accurate estimation about the object motion, which may make the operation unstable and hard to control in some cases. About the estimation of the mass and center of gravity of an unknown object, some researches have been carried on by firmly grasping and operating an object in robot hand [1] [2] [3], and by holding up and handling an object with a space robot hand [4]. However, these researches did not consider how to estimate the mass and center of gravity of an unknown object when the object is unable to be grasped by a robot hand.

Providing that the weight and/or shape of an object is over the grasping capacity of a robot hand, this paper proposes a technique that can estimate the mass and center of gravity of the graspless and shape-unknown object. A plane called *Gravity Equi-Effect Plane* is first defined, which contains the center of gravity and a contact line where the object is in line-contact with an environment. If three or more than three orientation-different gravity equi-effect planes of an object are obtained, the center of gravity of the object can be estimated by the intersect point of the planes. For estimating the gravity equi-effect plane, *Tip Operation* by robot finger, which tips the object repeatedly, is proposed. Using the fingertip position and force in tip operations, an algorithm to estimate the gravity equi-effect plane is addressed. Then an algorithm to estimate the mass and center of gravity of the object is given by several independent gravity equi-effect planes. Lastly, experimental verification on the proposed approach is performed.

2 Gravity Equi-Effect Plane and Tip Operation

2.1 Problem

The discussion of this paper is on the bases that:

- The object is a shape-unknown rigid polyhedron. And the mass and center of gravity of the object are also unknown. The environment is a plane whose position and normal are unknown.
- A robot hand will touch the object with point contact (touching point hereafter).
- The object is graspless by a robot hand because its weight and/or shape is over the grasping capacity of the robot hand.
- When the object is tipped and leaned by a robot hand, the object will be in line (or 2 points) contact with the environment (contact line hereafter).
- There is enough friction between the object and environment so that their contact position can be kept not to move during an object operation. If there is not enough friction, it is conceivable, for example, dexterously putting one (or two) finger at the contact part between the object and the environment to avoid the object slipping on the environment (for example see [13] or Fig.2(b)).

On these bases, the purpose of this paper is to estimate the mass and center of gravity of the graspless and shape-unknown object by dexterous object manipulation on an environment.

2.2 Gravity Equi-Effect Plane

Under the condition that an external force acting on an object is momentally balancing the gravity of the object around an axis, there exists a plane containing the axis and the center of gravity (see Fig.1). If the

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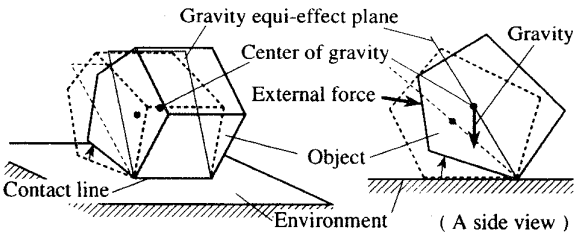


Fig.1 Gravity equi-effect plane

plane and the object have the same mass and the same position of the center of gravity, the external force acting on the object will have an equivalent effect on the plane. Hence, an object can be simplified as an equivalent plane when the moment effect around an axis is considered. The equivalent plane containing an axis and the center of gravity is referred to as *Gravity Equi-Effect Plane* in this paper.

The different axes on an object correspond to different gravity equi-effect planes. Three or more than three orientation-different gravity equi-effect planes intersect at one point, i.e., the center of gravity of the object. Hence, the center of gravity of object can be estimated by estimating three or more than three orientation-different gravity equi-effect planes of an object.

2.3 Tip Operation by Robot

In this section, we will give a dexterous object operation with robot finger, called *Tip Operation*, to obtain a number of fingertip positions and forces for estimating a gravity equi-effect plane.

Since the object is beyond robot hand's grasp, the contact with the environment will be used also for operating the object. The contact condition between the object and an environment plane can be classified into point contact, line contact and plane contact (see Fig.2). For the graspless object, the point contact is unstable and will be easy to change to a line contact or a plane contact. This paper, therefore, will deal with the line contact and plane contact to operate the object. The contact line of the line contact will be treated as an axis of a gravity equi-effect plane.

When an object is leaned at rest and is in line-contact with an environment (see Fig.3), the equilibrium condition of the moment about the contact line can be expressed as

$$\boldsymbol{l}_R \times \boldsymbol{f} + \boldsymbol{T} \times \boldsymbol{g} = \mathbf{o}, \tag{1}$$

$$\boldsymbol{T} \triangleq m\boldsymbol{l}_g, \tag{2}$$

where $\boldsymbol{l}_R$ is a distance vector from the contact line to

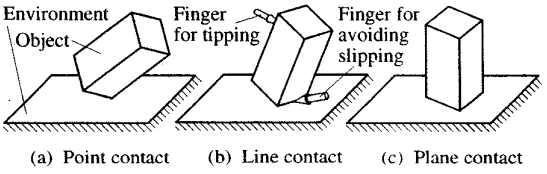


Fig.2 Three kind of contact conditions

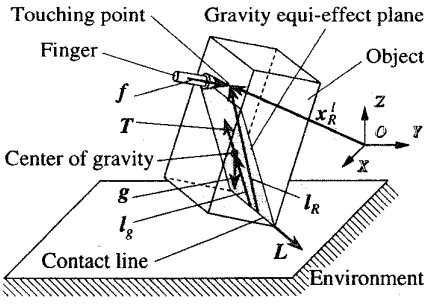


Fig.3 A leaned posture

the fingertip which keeps in touch with the object, $\boldsymbol{f}$ is the contact force at the fingertip measured by finger force sensor, $\boldsymbol{g}$ is the gravity. And $\boldsymbol{T}$ is defined as such a vector that crosses the contact line at right angles and passes through (or points at) the center of gravity (LPCG vector hereafter), m is the mass of the object, $\boldsymbol{l}_g$ is a distance vector from the contact line to the center of gravity. $\boldsymbol{l}_g$, m and $\boldsymbol{l}_R$ are unknown since the center of gravity, mass and shape of the object are unknown. Hence the LPCG vector $\boldsymbol{T}$ is also unknown.

As mentioned above, a gravity equi-effect plane contains an LPCG vector $\boldsymbol{T}$ and a contact line, which meet at right angles. If we know the LPCG vector $\boldsymbol{T}$ and the contact line, we can determine the gravity equi-effect plane. According to eq.(1), the unknown vector $\boldsymbol{l}_R$ must be estimated first for getting the vector $\boldsymbol{T}$. If there is enough friction between the object and environment, the contact line will not displace while the object is suitably leaned and turned around the contact line by a robot finger. Moving the object with this condition, a set of fingertip displacement and force information when the finger is tipping the object can be measured. In the next section, we will show that the vectors $\boldsymbol{l}_R$, $\boldsymbol{T}$ and contact line can be estimated from the measured fingertip displacement and force information. For estimating a gravity equi-effect plane, therefore, several independent fingertip displacements and fingertip contact forces should be obtained firstly.

To obtain the necessary information, we consider using robot finger to tip the object repeatedly at different fingertip positions under the condition that the contact line between the object and environment is kept not displacing. This object operation is referred to as *Tip Operation* in this paper (see Fig.4).

The procedure of tip operations can be set as:

Step 1 Set the object posture before leaned as an initial object posture when the object is in plane-contact with the environment. From the initial posture, use robot finger to lean the object around a contact line which is kept not displacing.

Step 2 Measure the force and position of fingertip when the object is kept in a leaned and still posture (leaned posture hereafter).

Step 3 Make the object return to the initial posture with keeping the current contact line not displacing. Then measure the fingertip position on the object which was touched by the fingertip at Step 2.

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Step 4 With changing the touching point position of the fingertip on the object, repeat Step 1 to Step 3 n ($n \geq 2$) times about the same contact line, where 2 or more than 2 touching points do not exist on a straight line which is parallel to the contact line.

On performing these steps, the force values of finger sensor will also be checked to avoid tumbling down the object. For example, if the value is less than the half of the value when a finger just starts to tip an object, a judgement can be given to stop tipping the object.

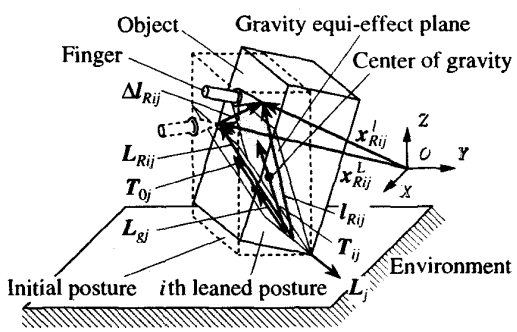
3 Estimation of Gravity Equi-Effect Plane

This section estimates the contact line and LPCG vector $\boldsymbol{T}$, then obtains the gravity equi-effect plane.

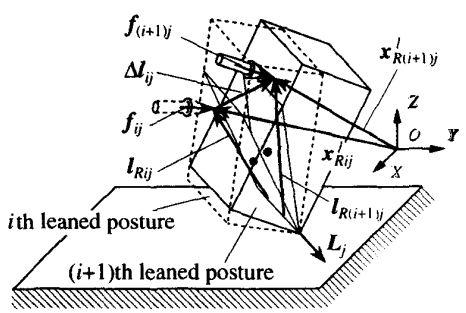
3.1 Vector from Contact Line to Touching Point

At first, let us consider estimating a vector which crosses a contact line at right angles and is from the contact line to a touching point of fingertip (LTTP vector hereafter).

Consider the relative positions among an object, an environment and a robot finger as shown in Fig.4. Let $O-XYZ$ denote the base frame on the environment, $C-UVW$ denote the object frame on the object. The following discussion will be based on the frame $O-XYZ$. Around a contact line j, let $\boldsymbol{L}_{Rij} \in \boldsymbol{R}^3$, $\boldsymbol{l}_{Rij} \in \boldsymbol{R}^3$ denote two LTTP vectors respectively for an initial object posture and a leaned object posture of ith tip operation, $\Delta\boldsymbol{l}_{Rij} \in \boldsymbol{R}^3$ denote ith fingertip displacement from the initial object posture to ith leaned object



(a) Initial posture and ith leaned posture



(b) ith leaned posture and $(i+1)$th leaned posture

Fig.4 Tip operations

posture, and let $\Delta\boldsymbol{l}_{ij} \in \boldsymbol{R}^3$ denote the distance vector of fingertip positions from ith leaned object posture to $(i+1)$th leaned object posture. Note that the angle between $\Delta\boldsymbol{l}_{Rij}$ and $\Delta\boldsymbol{l}_{R(i+1)j}$ should be large enough (for example, larger than $15^\circ \sim 20^\circ$) for making estimated errors as small as possible.

For ith and $(i+1)$th tip operations, we have

$$\boldsymbol{l}_{Rij} - \boldsymbol{L}_{Rij} = \Delta\boldsymbol{l}_{Rij}, \tag{3}$$

$$\boldsymbol{l}_{R(i+1)j} - \boldsymbol{l}_{Rij} = \Delta\boldsymbol{l}_{ij} - \frac{[\Delta\boldsymbol{l}_{Rij} \times \Delta\boldsymbol{l}_{R(i+1)j}]^T}{\|\Delta\boldsymbol{l}_{Rij} \times \Delta\boldsymbol{l}_{R(i+1)j}\|}\Delta\boldsymbol{l}_{ij}\frac{[\Delta\boldsymbol{l}_{Rij} \times \Delta\boldsymbol{l}_{R(i+1)j}]}{\|\Delta\boldsymbol{l}_{Rij} \times \Delta\boldsymbol{l}_{R(i+1)j}\|}, \tag{4}$$

where $\|*\|$ denotes the norm of vector $*$. Also, since LTTP vectors $\boldsymbol{l}_{Rij}$, $\boldsymbol{L}_{Rij}$ cross contact line j at right angles and have same magnitude, we can get

$$[\Delta\boldsymbol{l}_{Rij} \times \Delta\boldsymbol{l}_{R(i+1)j}]^T\boldsymbol{l}_{Rij} = 0, \tag{5}$$

$$[\Delta\boldsymbol{l}_{Rij} \times \Delta\boldsymbol{l}_{R(i+1)j}]^T\boldsymbol{L}_{Rij} = 0, \tag{6}$$

$$\boldsymbol{l}_{Rij}^T\boldsymbol{l}_{Rij} - \boldsymbol{L}_{Rij}^T\boldsymbol{L}_{Rij} = 0. \tag{7}$$

By using eq.(3), eq.(7) can be rewritten as

$$-2\Delta\boldsymbol{l}_{Rij}^T\boldsymbol{L}_{Rij} = \Delta\boldsymbol{l}_{Rij}^T\Delta\boldsymbol{l}_{Rij}. \tag{8}$$

For the case where the tip operation is repeated n times around contact line j with different touching points, we can define the following matrices:

$$\boldsymbol{C}_{1j} \triangleq [\ \boldsymbol{I}_{3n}\ \ -\boldsymbol{I}_{3n}\ ] \in \boldsymbol{R}^{3n\times 6n}, \tag{9}$$

$$\boldsymbol{C}_{2j} \triangleq [\boldsymbol{A}_{2j}\ \ \boldsymbol{O}_{3(n-1)\times 3n}] \in \boldsymbol{R}^{3(n-1)\times 6n}, \tag{10}$$

$$\boldsymbol{A}_{2j} \triangleq \mathrm{diag}\,[\,[-\boldsymbol{I}_3\ \ \boldsymbol{I}_3],\cdots,[-\boldsymbol{I}_3\ \ \boldsymbol{I}_3]\,] \in \boldsymbol{R}^{3(n-1)\times 3n}, \tag{11}$$

$$\boldsymbol{C}_{3j} \triangleq \mathrm{diag}\,[\boldsymbol{A}_{3\{1j\}}^T, \boldsymbol{A}_{3\{2j\}}^T, \cdots, \boldsymbol{A}_{3\{(n-1)j\}}^T, \boldsymbol{A}_{3\{(n-1)j\}}^T, \boldsymbol{A}_{3\{1j\}}^T, \boldsymbol{A}_{3\{2j\}}^T, \cdots, \boldsymbol{A}_{3\{(n-1)j\}}^T, \boldsymbol{A}_{3\{(n-1)j\}}^T] \in \boldsymbol{R}^{2n\times 6n}, \tag{12}$$

$$\boldsymbol{A}_{3\{ij\}} \triangleq [\Delta\boldsymbol{l}_{Rij} \times \Delta\boldsymbol{l}_{R(i+1)j}] \in \boldsymbol{R}^3, \tag{13}$$

$$\boldsymbol{C}_{4j} \triangleq [\boldsymbol{O}_{n\times 3n}\ \ -2\boldsymbol{A}_{4j}] \in \boldsymbol{R}^{n\times 6n}, \tag{14}$$

$$\boldsymbol{A}_{4j} \triangleq \mathrm{diag}\,[\Delta\boldsymbol{l}_{R1j}^T, \cdots, \Delta\boldsymbol{l}_{Rnj}^T] \in \boldsymbol{R}^{n\times 3n}, \tag{15}$$

$$\boldsymbol{c}_{1j} \triangleq [\Delta\boldsymbol{l}_{R1j}^T\ \ \cdots\ \ \Delta\boldsymbol{l}_{Rnj}^T]^T \in \boldsymbol{R}^{3n}, \tag{16}$$

$$\boldsymbol{c}_{2j} \triangleq [\boldsymbol{a}_{2\{1j\}}\ \ \cdots\ \ \boldsymbol{a}_{2\{(n-1)j\}}]^T \in \boldsymbol{R}^{3(n-1)}, \tag{17}$$

$$\boldsymbol{a}_{2\{ij\}} \triangleq \Delta\boldsymbol{l}_{ij} - \frac{\boldsymbol{A}_{3\{ij\}}^T}{\|\boldsymbol{A}_{3\{ij\}}\|}\Delta\boldsymbol{l}_{ij}\frac{\boldsymbol{A}_{3\{ij\}}}{\|\boldsymbol{A}_{3\{ij\}}\|} \in \boldsymbol{R}^3, \tag{18}$$

$$\boldsymbol{c}_{3j} \triangleq \boldsymbol{O}_{2n\times 1} \in \boldsymbol{R}^{2n}, \tag{19}$$

$$\boldsymbol{c}_{4j} \triangleq [\ \Delta\boldsymbol{l}_{R1j}^T\Delta\boldsymbol{l}_{R1j}\ \ \cdots\ \ \Delta\boldsymbol{l}_{Rnj}^T\Delta\boldsymbol{l}_{Rnj}\ ]^T \in \boldsymbol{R}^n. \tag{20}$$

By eqs.(3) (4) (5) (6) (8) and the above defined matrices, we can get

$$\boldsymbol{C}_j\boldsymbol{l}_j = \boldsymbol{c}_j \in \boldsymbol{R}^{3(3n-1)}, \tag{21}$$

$$\boldsymbol{C}_j \triangleq [\ \boldsymbol{C}_{1j}^T\ \ \boldsymbol{C}_{2j}^T\ \ \boldsymbol{C}_{3j}^T\ \ \boldsymbol{C}_{4j}^T\ ]^T \in \boldsymbol{R}^{3(3n-1)\times 6n}, \tag{22}$$

$$\boldsymbol{l}_j \triangleq [\ \boldsymbol{l}_{R1j}^T\ \ \cdots\ \ \boldsymbol{l}_{Rnj}^T\ \ \boldsymbol{L}_{R1j}^T\ \ \cdots\ \ \boldsymbol{L}_{Rnj}^T\ ]^T \in \boldsymbol{R}^{6n}, \tag{23}$$

$$\boldsymbol{c}_j \triangleq [\ \boldsymbol{c}_{1j}^T\ \ \boldsymbol{c}_{2j}^T\ \ \boldsymbol{c}_{3j}^T\ \ \boldsymbol{c}_{4j}^T\ ]^T \in \boldsymbol{R}^{3(3n-1)}, \tag{24}$$

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where $\boldsymbol{I}_*$ denotes a $*$-dimensional identity matrix and $\boldsymbol{O}_{*_1\times *_2}$ denotes a $(*_1\times *_2)$ zero matrix. When $n\geq 2$ and $\mathrm{rank}\boldsymbol{C}_j = 6n$, the unique solution (if no measured error) or the optimal approximate solution (if measured errors exist) to $\boldsymbol{l}_j$ can be obtained from eq.(21) since $3(3n-1) > 6n$. Therefore, considering that some measured errors can exist in $\Delta\boldsymbol{l}_{Rij}$ and $\Delta\boldsymbol{l}_{ij}$, the LTTP vectors $\boldsymbol{l}_{Rij}$, $\boldsymbol{L}_{Rij}$ $(i=1,2,\cdots,n,\ n\geq 2)$ for contact line j can be estimated by

$$\boldsymbol{l}_j = \boldsymbol{C}_j^{+}\boldsymbol{c}_j, \tag{25}$$

where $\boldsymbol{C}_j^{+}\in \boldsymbol{R}^{6n\times 3(3n-1)}$ is the pseudoinverse of $\boldsymbol{C}_j$.

3.2 Estimation of Contact Line

Since $\Delta\boldsymbol{l}_{Rij}$ and contact line j cross at right angles, we have

$$[\Delta\boldsymbol{l}_{Rij}\times\Delta\boldsymbol{l}_{R(i+1)j}]^T\boldsymbol{L}_{ij} = k_{Sij}\|\Delta\boldsymbol{l}_{Rij}\times\Delta\boldsymbol{l}_{R(i+1)j}\|, \tag{26}$$

$$k_{Sij}\triangleq \mathrm{sgn}\left\{[\Delta\boldsymbol{l}_{Rij}\times\boldsymbol{g}]^T\left[\Delta\boldsymbol{l}_{Rij}\times\Delta\boldsymbol{l}_{R(i+1)j}\right]\right\}, \tag{27}$$

where $\boldsymbol{L}_{ij}\in\boldsymbol{R}^3$ represents the unit vector of contact line j to ith tip operation, whose direction is given by $[\Delta\boldsymbol{l}_{Rij}\times\boldsymbol{g}]^T\boldsymbol{L}_{ij} > 0$. Because $\boldsymbol{L}_{ij}$ is parallel to $\Delta\boldsymbol{l}_{Rij}\times\Delta\boldsymbol{l}_{R(i+1)j}$, $\boldsymbol{L}_{ij}$ has the minimum norm in the vectors satisfying eq.(26). After repeated the tip operation $n(\geq 2)$ times about contact line j, there are

$$\boldsymbol{D}_{Lj}\widehat{\boldsymbol{L}}_j = \boldsymbol{d}_{Lj}\in\boldsymbol{R}^{(n-1)}, \tag{28}$$

$$\boldsymbol{D}_{Lj}\triangleq \mathrm{diag}\left[\ [\Delta\boldsymbol{l}_{R1j}\times\Delta\boldsymbol{l}_{R2j}]^T,\cdots,[\Delta\boldsymbol{l}_{R(n-1)j}\times\Delta\boldsymbol{l}_{Rnj}]^T\right] \in\boldsymbol{R}^{(n-1)\times 3(n-1)}, \tag{29}$$

$$\widehat{\boldsymbol{L}}_j\triangleq\left[\ \boldsymbol{L}_{1j}^T\ \cdots\ \boldsymbol{L}_{(n-1)j}^T\ \right]\in\boldsymbol{R}^{3(n-1)}, \tag{30}$$

$$\boldsymbol{d}_{Lj}\triangleq\left[\ K_{S1j}\|\Delta\boldsymbol{l}_{R1j}\times\Delta\boldsymbol{l}_{R2j}\|\ \cdots\ K_{S(n-1)j}\|\Delta\boldsymbol{l}_{R(n-1)j}\times\Delta\boldsymbol{l}_{Rnj}\|\ \right]^T\in\boldsymbol{R}^{(n-1)}. \tag{31}$$

The solution with minimum norm of eq.(28) can be obtained by using the pseudoinverse of $\boldsymbol{D}_{Lj}$. Thereby, considering that some measured errors can exist in $\Delta\boldsymbol{l}_{Rij}$, the direction $\boldsymbol{L}_j$ of contact line j can be estimated by

$$\boldsymbol{L}_j = \frac{\boldsymbol{E}_{Lj}\widehat{\boldsymbol{L}}_j}{\|\boldsymbol{E}_j\widehat{\boldsymbol{L}}_j\|}\in\boldsymbol{R}^3,\qquad \|\boldsymbol{L}_j\| = 1, \tag{32}$$

$$\widehat{\boldsymbol{L}}_j = \boldsymbol{D}_{Lj}^{+}\boldsymbol{d}_{Lj}\in\boldsymbol{R}^{3(n-1)}, \tag{33}$$

$$\boldsymbol{E}_{Lj}\triangleq[\ \boldsymbol{I}_3\ \cdots\ \boldsymbol{I}_3\ ]\in\boldsymbol{R}^{3\times 3(n-1)}, \tag{34}$$

where $\boldsymbol{D}_{Lj}^{+}\in\boldsymbol{R}^{3(n-1)\times(n-1)}$ is the pseudoinverse of $\boldsymbol{D}_{Lj}$.

Using the obtained $\boldsymbol{L}_j$ and $\boldsymbol{l}_j$, we can estimate contact line j by

$$\boldsymbol{x} = \boldsymbol{p}_{Lj} + k_{Lj}\boldsymbol{L}_j\in\boldsymbol{R}^3,\qquad \|\boldsymbol{L}_j\| = 1, \tag{35}$$

$$\boldsymbol{p}_{Lj}\triangleq\frac{1}{2n}\left\{\sum_{i=1}^{n}\left(\boldsymbol{x}_{Rij}^{l}-\boldsymbol{L}_j^T\boldsymbol{x}_{Rij}^{l}\boldsymbol{L}_j+\boldsymbol{l}_{Rij}\right) +\sum_{i=1}^{n}\left(\boldsymbol{x}_{Rij}^{L}-\boldsymbol{L}_j^T\boldsymbol{x}_{Rij}^{L}\boldsymbol{L}_j+\boldsymbol{L}_{Rij}\right)\right\}, \tag{36}$$

where $\boldsymbol{p}_{Lj}\in\boldsymbol{R}^3$ is a representative point on contact line j, which is estimated by the arithmetical mean since some measured errors can exist. k_{Lj} denotes the distance form the representative point along the line. And $\boldsymbol{x}_{Rij}^{l}$, $\boldsymbol{x}_{Rij}^{L}$ denote the touching point positions from the origin of $O-XYZ$ with ith initial object posture and ith leaned object posture respectively (see Fig.4(a)).

3.3 Estimation of LPCG Vector

Now let us consider estimating the LPCG vector $\boldsymbol{T}_j$ according to the estimated $\boldsymbol{l}_j$ and the fingertip forces $\boldsymbol{f}_{ij}$ $(i=1,2,\cdots,n,\ n\geq 2)$, which are measured when the object is on leaned postures.

By repeating the tip operation $n(\geq 2)$ times, the following matrices can be derived from eq.(1),

$$\boldsymbol{G}_{1j}\boldsymbol{T}_j = \boldsymbol{b}_{1j}\in\boldsymbol{R}^n, \tag{37}$$

$$\boldsymbol{G}_{1j}\triangleq[\ \boldsymbol{O}_{3n\times 3}\ \ \boldsymbol{B}_{1j}\ ]\in\boldsymbol{R}^{3n\times 3n}, \tag{38}$$

$$\boldsymbol{B}_{1j}\triangleq\mathrm{diag}[\ -[\boldsymbol{g}\times],\cdots,-[\boldsymbol{g}\times]\ ]\in\boldsymbol{R}^{3n\times 3(n-1)}, \tag{39}$$

$$\boldsymbol{T}_j\triangleq\left[\ \boldsymbol{T}_{0j}^T\ \ \boldsymbol{T}_{1j}^T\ \cdots\ \boldsymbol{T}_{nj}^T\ \right]^T\in\boldsymbol{R}^{3(n+1)}, \tag{40}$$

$$\boldsymbol{b}_{1j}\triangleq\left[\ [\boldsymbol{l}_{R1j}\times\boldsymbol{f}_{1j}]^T\ \cdots\ [\boldsymbol{l}_{Rnj}\times\boldsymbol{f}_{nj}]^T\right]^T\in\boldsymbol{R}^{3n}, \tag{41}$$

where $[\boldsymbol{g}\times]$ represents a skew symmetry matrix defined by $[\boldsymbol{g}\times]\boldsymbol{l}_{Rij}\triangleq\boldsymbol{g}\times\boldsymbol{l}_{Rij}$, $\boldsymbol{T}_{0j}$ and $\boldsymbol{T}_{ij}$ correspond the initial object posture and ith leaned object posture respectively.

When the object is leaned and turned about contact line j with ith tip operation, the angle between LPCG vectors $\boldsymbol{T}_{ij}$ and $\boldsymbol{T}_{0j}$ is the same as the angle between LTTP vectors $\boldsymbol{l}_{Rij}$ and $\boldsymbol{L}_{Rij}$, and the magnitudes of $\boldsymbol{T}_{ij}$ and $\boldsymbol{T}_{0j}$ are the same. Accordingly, there are

$$\boldsymbol{T}_{0j}-\boldsymbol{R}_{Rij}\boldsymbol{T}_{ij} = \boldsymbol{O}_{3\times 1}\in\boldsymbol{R}^3, \tag{42}$$

$$\boldsymbol{R}_{Rij}\triangleq\boldsymbol{r}_{ij}\boldsymbol{R}_{ij}\boldsymbol{r}_{ij}^T\in\boldsymbol{R}^{3\times 3}, \tag{43}$$

$$\boldsymbol{r}_{ij}\triangleq\left[\ \frac{\boldsymbol{L}_{Rij}\times\boldsymbol{L}_j}{\|\boldsymbol{L}_{Rij}\times\boldsymbol{L}_j\|}\ \ \frac{\boldsymbol{L}_{Rij}}{\|\boldsymbol{L}_{Rij}\|}\ \ \boldsymbol{L}_j\ \right]\in\boldsymbol{R}^{3\times 3}, \tag{44}$$

$$\boldsymbol{R}_{ij}\triangleq\begin{bmatrix}\dfrac{\boldsymbol{L}_{Rij}^T\boldsymbol{l}_{Rij}}{\|\boldsymbol{L}_{Rij}\|\|\boldsymbol{l}_{Rij}\|} & -\dfrac{\|\boldsymbol{L}_{Rij}\times\boldsymbol{l}_{Rij}\|}{\|\boldsymbol{L}_{Rij}\|\|\boldsymbol{l}_{Rij}\|} & 0\\ \dfrac{\|\boldsymbol{L}_{Rij}\times\boldsymbol{l}_{Rij}\|}{\|\boldsymbol{L}_{Rij}\|\|\boldsymbol{l}_{Rij}\|} & \dfrac{\boldsymbol{L}_{Rij}^T\boldsymbol{l}_{Rij}}{\|\boldsymbol{L}_{Rij}\|\|\boldsymbol{l}_{Rij}\|} & 0\\ 0 & 0 & 1\end{bmatrix}\in\boldsymbol{R}^{3\times 3} \tag{45}$$

Repeating the tip operation $n(\geq 2)$ times, we have

$$\boldsymbol{G}_{2j}\boldsymbol{T}_j = \boldsymbol{O}_{3n\times 1}\in\boldsymbol{R}^{3n}, \tag{46}$$

$$\boldsymbol{G}_{2j}\triangleq[\ \boldsymbol{E}_{2j}\ \ -\boldsymbol{B}_{2j}\ ]\in\boldsymbol{R}^{3n\times 3(n+1)}, \tag{47}$$

$$\boldsymbol{B}_{2j}\triangleq\mathrm{diag}\left[\ \boldsymbol{R}_{R1j}^T,\cdots,\boldsymbol{R}_{Rnj}^T\ \right]\in\boldsymbol{R}^{3n\times 3n}, \tag{48}$$

$$\boldsymbol{E}_{2j}\triangleq[\ \boldsymbol{I}_3\ \cdots\ \boldsymbol{I}_3\ ]^T\in\boldsymbol{R}^{3n\times 3}. \tag{49}$$

By eqs.(37) and (46), we can get

$$\boldsymbol{G}_j\boldsymbol{T}_j = \boldsymbol{b}_j\in\boldsymbol{R}^{6n}, \tag{50}$$

$$\boldsymbol{G}_j\triangleq\left[\ \boldsymbol{G}_{1j}^T\ \ \boldsymbol{G}_{2j}^T\ \right]^T\in\boldsymbol{R}^{6n\times 3(n+1)}, \tag{51}$$

$$\boldsymbol{b}_j\triangleq\left[\ \boldsymbol{b}_{1j}^T\ \ \boldsymbol{O}_{3n\times 1}^T\ \right]\in\boldsymbol{R}^{6n}. \tag{52}$$

Same as the solution to $\boldsymbol{l}_j$, when $n\geq 2$ and $\mathrm{rank}\boldsymbol{G}_j = 3(n+1)$, the optimal approximate solution of $\boldsymbol{T}_j$ can be obtained from eq.(50) since $6n > 3(n+1)$. Accordingly,

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considering that some measured errors can exist, the vector $\boldsymbol{T}_j$ can be estimated by

$$\boldsymbol{T}_j = \boldsymbol{G}_j^+ \boldsymbol{b}_j, \tag{53}$$

where $\boldsymbol{G}_j^+ \in \boldsymbol{R}^{3(n+1)\times 6n}$ is the pseudoinverse of $\boldsymbol{G}_j$.

3.4 Estimation of Gravity Equi-Effect Plane

From the contact line $\boldsymbol{x} = \boldsymbol{p}_{Lj} + k_{Lj}\boldsymbol{L}_j$ and the LPCG vector $\boldsymbol{T}_{0j}$, we can estimate the gravity equi-effect plane corresponding contact line j as

$$[\boldsymbol{L}_j \times \boldsymbol{T}_{0j}]^T(\boldsymbol{x} - \boldsymbol{p}_{Lj}) = 0. \tag{54}$$

As the above discussion, by performing the tip operation to the different contact line between the object and environment with the same initial object posture, the different gravity equi-effect planes can be estimated respectively.

4 Estimation of Center of Gravity and Mass

4.1 Estimation of Center of Gravity

Since the obtained gravity equi-effect planes ($j = 1, 2, \cdots, k,\ k \geq 3$) all contain the center of gravity as mentioned previously, we have

$$\boldsymbol{H}\boldsymbol{P}_g = \boldsymbol{h} \in \boldsymbol{R}^k, \qquad k \geq 3, \tag{55}$$

$$\boldsymbol{H} \triangleq [\,[\boldsymbol{L}_1 \times \boldsymbol{T}_{01}] \ \cdots \ [\boldsymbol{L}_k \times \boldsymbol{T}_{0k}]\,]^T \in \boldsymbol{R}^{k\times 3}, \tag{56}$$

$$\boldsymbol{h} \triangleq [\,\boldsymbol{p}_{L1}^T[\boldsymbol{L}_1 \times \boldsymbol{T}_{01}] \ \cdots \ \boldsymbol{p}_{Lk}^T[\boldsymbol{L}_k \times \boldsymbol{T}_{0k}]\,]^T \in \boldsymbol{R}^k, \tag{57}$$

from eq.(54), where $\boldsymbol{P}_g \in \boldsymbol{R}^3$ denotes the position of the center of gravity. Note that the angle between two plane's normals should be large enough (for example, larger than 20 °) for making estimated errors as small as possible.

When $k \geq 3$, $\mathrm{rank}\boldsymbol{H} = 3$, the optimal approximate solution of $\boldsymbol{P}_g$ can be obtained from eq.(55). If an object is in plane-contact with an environment, the contact plane between the two polyhedrons will be a polygon. Hence for an initial object posture, at least three polygon edges can be used as three independent contact lines, so that $k \geq 3$ can be guaranteed. Accordingly, we can estimate the center of gravity of the object by

$$\boldsymbol{P}_g = \boldsymbol{H}^+\boldsymbol{h}, \tag{58}$$

where $\boldsymbol{H}^+ \in \boldsymbol{R}^{3\times k}$ is the pseudoinverse of $\boldsymbol{H}$.

4.2 Estimation of Mass of Object

From the obtained $\boldsymbol{P}_g$, a vector $\boldsymbol{l}_{gj}$, which crosses contact line j at right angles and is toward the center of gravity, can be described by

$$\boldsymbol{l}_{gj} = (\boldsymbol{P}_g - \boldsymbol{p}_{Lj}) - \boldsymbol{L}_j^T(\boldsymbol{P}_g - \boldsymbol{p}_{Lj})\boldsymbol{L}_j \in \boldsymbol{R}^3. \tag{59}$$

Considering that some measured errors can exist, from eq.(2) the object mass m can be estimated by

$$m = \frac{1}{k}\sum_{j=1}^{k}\frac{\|\boldsymbol{T}_{0j}\|}{\|\boldsymbol{l}_{gj}\|}, \qquad k \geq 3. \tag{60}$$

5 Verification by Experiment

This section demonstrates the validity of the proposed method by three experiments. The unknown object for the first and second experiments is a wooden box shown in Fig.5(a) and (b). For the second experiment, a weight is added on the bottom of the box in order to change the mass and center of gravity of the box. The unknown object of the third experiment is a shape-complex wooden table shown in Fig.5(c), whose center of gravity is difficult to know in general. The experimental environment is an unknown plane and there is enough friction between the objects and the environment plane. In the experiments, the contact lines between the objects and environment plane are kept not to move because of the enough friction, while the touching positions between the fingertip and objects are allowed to move. After an object is leaned and the touching position and force of the fingertip are measured, the touching position will be marked on the object. Then, when the object is returned to its initial posture, the marked point is positionally measured again by fingertip to get the touching position at initial posture, so that fingertip displacements Δl_{Rij} can be obtained.

The estimated results and the reference values about the centers of gravity and masses of the objects are shown in Table 1 to Table 3, where the centers of gravity are given with regard to $C - UVW$ for convenience. The reference values are obtained by human skill for the centers of gravity and a dial scale for masses. But the center of gravity of the table is unable to be measured by human skill because of its complex shape. Comparing the estimated and reference values, the relative errors of the position of the centers of gravity are less than 3%, and the relative errors of masses of the objects are less than 1%. Considering that the accuracy of force sensor is 2% and the position error of robot arm

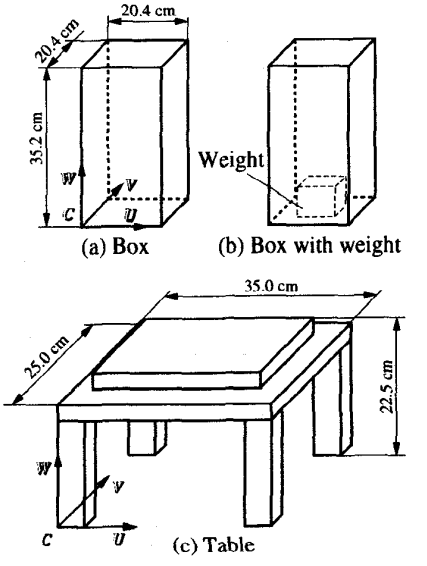


Fig.5 Experimental objects

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Table 1 Result of experiment 1

	P_{gu} [cm]	P_{gv} [cm]	P_{gw} [cm]	m [kg]
Reference value	10.20	10.40	17.95	1.872
Estimated value	10.28	10.18	17.42	1.888
Error (%)	0.8	2.1	2.9	0.85

Table 2 Result of experiment 2

	P_{gu} [cm]	P_{gv} [cm]	P_{gw} [cm]	m [kg]
Reference value	10.20	10.38	16.08	2.115
Estimated value	10.16	10.16	15.89	2.131
Error (%)	0.4	2.1	1.2	0.76

Table 3 Result of experiment 3

	P_{gu} [cm]	P_{gv} [cm]	P_{gw} [cm]	m [kg]
Reference value	-	-	-	1.771
Estimated value	17.81	12.50	18.52	1.780
Error (%)	-	-	-	0.51

is 1%, the estimated results have enough accuracy and the validity of proposed approach is shown.

6 Conclusion

In many cases for manipulating an object stably and accurately by robot, it is required to know the center of gravity and mass of the object. To the contrary, it will be difficulty to make an accurate estimation about the object motion, which may make the operation unstable and hard to control. Providing that the weight and/or shape of an object is over the grasping capacity of a robot hand, this paper has proposed a technique that can estimate the mass and center of gravity of the graspless and shape-unknown object.

A plane called *Gravity Equi-Effect Plane* was defined in this paper, which contains the center of gravity and a contact line where the object is in line-contact with an environment. Three or more than three orientation-different gravity equi-effect planes will intersect at one point, i.e., the center of gravity of the object. Therefore, the center of gravity of object can be estimated by estimating three or more than three orientation-different gravity equi-effect planes of an object. For estimating the gravity equi-effect plane, *Tip Operation* by robot finger, which tips the object repeatedly, was proposed. Using the fingertip position and force information measured from tip operations, an algorithm to estimate the gravity equi-effect plane was described. Then an algorithm to estimate the center of gravity and mass of the object was given by several independent gravity equi-effect planes. Lastly, experimental verification on the proposed approach was performed and its results were outlined.

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